

Adaptive Transfer Learning for Field-Based Lentil Disease Detection

Sai Nithin Chowdary Kota and Sourajit Behera^[0000–0002–2882–9115]

Dept. of Computer Science & Engineering, SVNIT Surat, India
{p24ds023,sourajit}@coed.svnit.ac.in

Abstract. Agriculture remains vital for human sustenance, yet the early detection of plant diseases remains a significant challenge; delays, even minor ones, can significantly reduce crop quality, yield, and productivity. Thanks to advances in deep learning and computer vision, automated plant disease detection has become a practical reality. However, most existing models still primarily focus on accuracy, often overlooking important practical concerns, such as efficiency and ease of deployment. Moreover, much of the existing research focuses on developing deep convolutional neural networks from scratch; this work leverages a pre-trained ResNet101 model through transfer learning and empirically fine-tunes its layers. This empirical fine-tuning addresses the limitations of prior studies, where frozen and trainable layers are typically hard-coded without empirical justification. We conducted experiments to determine the ideal fine-tuning depth that maintains a strong balance between accuracy and efficiency. Compared to other pre-trained CNN models, our approach achieves an overall performance improvement of approximately 4%, while maintaining comparable results across multiple disease categories. Notably, it outperforms others by 17% in precision and 3% in F1-score for the *Ascochyta Blight* class on a publicly available real-world lentil disease dataset. With roughly 5 million trainable parameters, the model achieves an effective balance between accuracy and efficiency, underscoring its suitability for real-world deployment.

Keywords: Lentil disease detection · Computer vision · Deep learning · Transfer learning · Convolutional Neural Networks

1 Introduction

For ages and years to come, agriculture has served as one of the fundamental backbones sustaining human civilization, providing the food basket and making it essential for economic resilience. Over a substantial period of time, spanning approximately 20 years from 2000 onwards, at the global level, the accumulated costs of agriculture, forestry, and fishing created a positive momentum of 78%, amassing around 3.6 trillion dollars [7]. However, agriculture is prone to facing several key challenges, with plant diseases being one of the key threats. Diseases can deteriorate the crop’s quality, reduce the yield quantity, and subsequently increase economic losses to farmers [16]. For example, concerning corns, gray

leaf spots materialize as yellow spots, which later expand to cover the leaf veins, forming a rectangular shape that is often gray and is largely mistaken for common rust. These gray leaf spots result in a staggering 25 million tonnes less in lost yields, making a significant dent of approximately 35% across North America.

Depending on the area and amount of production on a worldwide scale, lentils, a food legume, rank in the sixth position [1]. Because of the high micronutrient, fibre, protein, and mineral content, lentils remain a sought-after and prevalent legume in Bangladesh and worldwide [13]. Despite being cultivated in over 70 countries, the Indian subcontinent, Australia, America, and Africa produce around 70% of the total legumes [4]. According to FAOSTAT statistics (2020), India ranks second among the top ten countries producing lentils worldwide [8]. Among the multiple factors that hinder their production, the progression of diseases like *Stemphylium* blights has been reported in almost all countries that grow lentils. As a result, there was a significant decline in lentil production globally, amounting to a 22% decrease [10]. The disease led to substantial yield losses in major growing countries, such as India, by 93% [15], and Nepal by 100%, due to severe infestation in the sensitive lentil variant [9]. Therefore, the accurate identification and prompt recognition of disease signs are crucial for effective prevention and management. Nevertheless, farmers come across hiccups using this approach, as it can be labor-intensive while being highly prone to errors when making observations with the naked eye.

Recently, using machine learning (ML) approaches for solving and improving the disease detection problems in agriculture has been one of the prominent areas under constant attention. Earlier works utilized the plain versions of ML approaches to categorize leaf diseases via the standard two-staged pipeline: (a) create feature extraction techniques based on texture, color, and shape; (b) utilize the features grasped from (a) for training classification models. Detailing further, for example, authors in [2] presented an approach for extracting gray-level-co-occurrence attributes based on leaf texture on the PlantVillage dataset. Next, the features extracted were fed into a support vector machine (SVM) and attained an 83.7% accuracy on corn images. Shah et al. [14] obtained features based on six color channels and fed them into an artificial neural network (ANN). This approach used a simple model while requiring a small training image set. However, it demands the derivation of meaningful hand-crafted features that demand significant domain expertise, apart from depicting the chances of making the classification ineffective when dealing with complex data with assorted data distributions. Therefore, deep learning (DL) networks with their edge in automatic feature extraction and data handling have been the focus of many recent works, as they hold the key to overcoming the problems through self-learning-based meaningful learning of features.

Among DL-based networks, convolutional neural networks (CNNs) have been apprehended as one of the most successful approaches to serve computer vision problems. With plant disease datasets that utilize image inputs, CNNs have become the preferred choice for analysis. For example, to grasp a broad image feature applicable to various contexts, CNNs have been given both natural objects

and synthetic objects as inputs. With the number of such objects ranging into millions, multiple state-of-the-art datasets have been developed, including ImageNet [6], ILSRVC, or CIFAR-100. Trained using datasets like the above, CNNs often outperform in feature extraction, ingraining expressive feature recognition, and outclassing manual feature engineering, which demands significant domain expertise. Ashurov et al. [3] tested with the combination of improving residual skip connection along with squeeze & excitation blocks to design a modified depthwise CNN for plant disease detection. Despite their popularity, CNNs are often short-handed when low-data regime datasets are involved, making them incapable of revealing the issue [5]. For such problems, transfer learning (TL) can be beneficial by improving pre-trained deep CNNs to achieve better estimations while coping with data scarcity [18]. TL revolves around the reuse and retention policy of nonfluctuating general features captured in a source domain, aiming to enhance performance on a different but related problem in the target domain.

Teena et al. [17] performed a comparative study among several pre-trained models trained initially on the ImageNet dataset for mango disease classification. Such studies have confirmed the effectiveness of TL, yet the selection of fine-tuning remains a crucial factor governing overall performance. In cases of complicated tasks in the target domain, the training often needs to be performed on the whole network from pre-trained weights. However, for functions that are much closer to the source domain, retraining only a subset of layers can suffice. Thus, the extent of layer retraining holds the key to performance metrics. However, many earlier works have adapted fixed, hard-coded retraining schemes without empirical justification, which might be a crucial limitation of such works. Furthermore, pre-trained models have already been trained on large datasets with millions of images, making them inherently parameter-heavy, while boosting accuracy and increasing the time and computational cost. To achieve smart agriculture, approaches that enable flexible adjustments of trainable parameters will always be in demand, yet remain largely unexplored. To the best of the author's knowledge, few studies have empirically validated fine-tuning strategies and simultaneously reported parameter trade-offs. For overcoming the above, this study presents an experimental approach combining CNNs while investigating fine-tuning strategies, empirical validation of layer retraining, while reporting performance trade-offs and relevant trainable parameter counts. The work has been validated on a field-based, real-world lentil crop dataset with the primary contributions summed up as follows:

- A TL-based fine-tuning approach is proposed, which retrains a selected set of layers. Simulations help find out how many layers to retrain, aiming for balancing the performance with concern for parameter count, making it a preliminary step in the direction of edge deployment- a direction mostly left unaddressed.
- Simulations involve comparison among several pre-trained models apart from state-of-the-art work, classification report across disease classes, to highlight the model's benefits.

- Discussions on the deficiencies and conceivable enhancements of the model proposed have been undertaken.

The rest of the paper has been assembled as follows. Section 2 presents the background, followed by Section 3 that represents the proposed approach. Section 4 discusses the dataset, settings, results, and discussions. Finally, the conclusion and future work are covered in Section 5.

2 Materials & Methods

2.1 Convolutional Neural Networks (CNN)

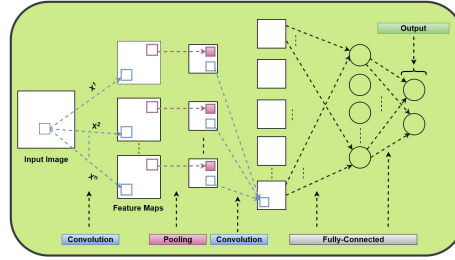


Fig. 1. Pictorial representation of a basic CNN

Being one of the most popular among feed-forward networks, CNNs excel at showcasing meaningful properties, such as weight sharing, subsampling, and local receptive fields. These properties have enhanced it to make it one of the most sought-after methods for working with $2D$ inputs. In this regard, we present Fig. 1, which depicts a standard CNN block comprising components such as convolutional layers, fully connected layers, and pooling layers used to build deep networks. Using kernels for the convolution of inputs, a convolutional layer can extract a set of outputs. With kernels of identical size, features get grasped at the local level under each input region. Consider a matrix of input features R fed through an activation function $!$ to a model comprising a set of U stacked layers. Employing a kernel X of $m \times m$, the k^{th} component of the B^{th} convolutional layer R_k^B totaled up with a bias L_k^B can be expressed as:

$$R_k^B = ! \left(\sum_{D \in D_k} (R_k^{B-1} \times U_{D_k}^B) + L_k^B \right) \quad (1)$$

Where $\times$ exhibits the convolutional operation, R_k^{B-1} the k^{th} of $(B-1)^{th}$ layer and D_k the input feature matrix used to compute the k^{th} consequence. Following the convolutional layer is the pooling layer that reduces the spatial dimensional proportions & parameters to ensure translational invariance. With

$o(.)$ specifying a sub-sampling function and α a multiplicative bias, the k^{th} component of the B^{th} pooling layer R_k^B can be expressed as:

$$R_k^B = (\alpha_k^B o(R_k^{B-1}) + L_k^B) \quad (2)$$

The $o(.)$ function can perform one of the following selections: max, global, or average. These operations are generally aggregated across each $m \times m$ different section input to reduce the outcome by a factor of m along both spatial dimensions. Each subspace exemplifies the most significant value to display the output entry. In the case of average pooling, the average operation on each subspace operates as the output. Next, the global pooling decreases the full feature map into a single value instead of the input map's downsampling spaces. Finally, fully connected layers help to collect high-level patterns from earlier layers for estimating the instance probability from the test set, mapping to class sets fed using an activation function. With P^B presenting the weights among $(B-1)^{th}$ and B^{th} layer, the output of fully connected B^{th} layer, R^B can be presented as:

$$R^B = ((P^B)^T R^{B-1} + L^B) \quad (3)$$

2.2 Transfer Learning and Fine-Tuning

In general, ML approaches are trained on a specific characteristic for solving typical problems. Thus, the corresponding approach might underperform whenever modifications occur in data distributions or new jobs are presented. Solving such issues may suggest retraining from scratch as a method. TL utilizes models that have already been trained to solve a similar task in a domain (for example, s), which are then used later on with reduced training to solve another issue in another domain (for example, t). With the widespread use of deep networks, the pre-trained parameters from domain s can revitalize in solving a related problem in domain t with minuscule computation & training. For example, when a highly complex task is involved, training entire networks using saved weights is beneficial. Yet, focusing on training of a few dense or convolutional layers often suffices for comparable tasks.

3 Proposed Methodology

Fig. 2 specifies the overall representation of our proposed approach for lentil disease detection. The depiction exclusively uses lock symbols to represent fine-tuning and retraining specific layers. The pre-trained CNN model that achieves the best results across multiple performance metrics is ResNet101 (readers are referred to Section 4.3 and Table 3 for further details). Subsequently, the base ResNet-101 has been followed by a dense block of layers whose configurations have been provided in Section 4.2. Finally, the last dense layer predicts the output of the given input image. Once all the pictures in the test set have been exhausted, the robustness of the method is evaluated using the performance metrics. During the training phase, the cross-entropy function serves as the cost function.

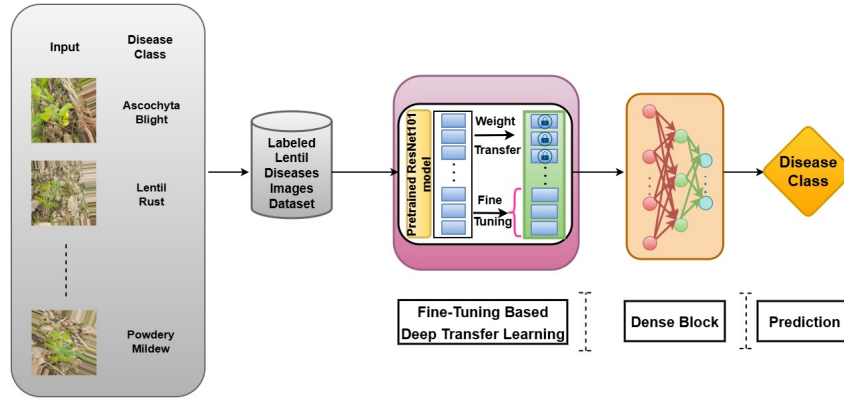


Fig. 2. Expanded depiction of the proposed approach, where locks suggest frozen layers, with fine-tuning examined empirically

4 Experiments, Results, & Discussions

4.1 Dataset Description

The data used in this work have been taken from [11] where image data were captured at various stages and situations of the lentil crop, together with the plants that were healthy, alongside those impacted by diseases such as powdery mildew, lentil rust, and ascochyta blight. The image data have been procured from farms by capturing them using a smartphone between May and August 2023 in Bangladesh. With each image saved using the JPG format and having a resolution of 224×224 pixels, labeling has been provided corresponding to each disease category. The images were apprehended during wee hours, specifically between 6 AM & 9 AM and 4 PM & 6 PM for reducing shadows by benefiting from natural lighting. Although it can help reduce extreme variability, it can't fully guarantee a difference in luminance due to seasonal or cloudy environments, making it partially controlled in settings. Efforts were made for avoidance of harsh sunlight as it could cause glare. Despite the adherence to usage in consistent lighting to reduce shadow impact, there exists a possibility of varying sunlight intensity and shadows daily, making it an uncontrolled setting. Marked by warm temperatures and infrequent rain, during the early months of early summer and late spring, the circumstances aided in the healthy growth of lentils and observing different symptoms of diseases. As the weather (in terms of humidity and temperature here) cannot be subjected to standardization, it is susceptible to natural deviations, making the dataset uncontrolled, as it has been compounded by field-based data collection rather than laboratory-based. The smartphone utilized to gather data had a 64-megapixel camera featuring a GW1 sensor, measuring 1/1.72 inches in size, with respective pixels measuring around 0.8 micrometers. Comprising four categories: Normal, Lentil rust, Ascochyta blight, and Powdery mildew, the original image count consisted of

1,898 images in total, which was later subjected to data augmentation, taking the tally to 4,550. The distribution of pictures under each category is presented in Table 1.

Table 1. Dataset Description

Disease Category	Images in Original data	Images in Augmented data
Normal	454	1321
Ascochyta Blight	466	1029
Powdery Mildew	492	1125
Lentil Rust	480	1075
Total	1898	4550

4.2 Settings

Simulations have been conducted to estimate the various disease categories affecting the lentil crop. We have used the augmented dataset version (refer to Table 1) for the training and testing evaluations in this work. Our proposed pre-trained CNN model, ResNet101, has been subjected to empirical layer re-training and freezing for training on the lentil dataset. With a TensorFlow backend and Keras library, we utilized the functional API to implement the network, while constantly monitoring the validation loss to achieve practical training. As the dataset already has dedicated folders for training, validation, and testing images, no explicit splitting has been applied in our implementation. During implementation, apart from the base ResNet101, we introduced a set of successive layers, including, in order, a global average pooling layer, a dropout layer to mitigate the chances of overfitting, and a dense layer with 512 neurons. The dense layer was configured with the ReLU activation with an L2 regularizer. Finally, we placed a dense layer with the same number of neurons as the number of disease categories, which was then subjected to the softmax activation. Concerning the hyperparameters, we present them in Table 2. The network and all experiments were conducted using a Tesla P100 NVIDIA GPU with 16 GB RAM on Colab or a Tesla T4 with 16 GB VRAM on a Kaggle notebook, depending on the availability of resources. All images in the dataset have been fixed at 224×224 . After training, the trained model was evaluated on the test set, and performance metrics were presented.

4.3 Performance Metrics

The choice of specific performance metrics hinges on the explicit objectives of the focused method. While accuracy has been one of the most recognized metrics, holding it in isolation can be deceptive when dealing with imbalanced [12] and diverse data distributions. In our study, significant emphasis has been placed on macro-averaging, often referred to as balanced accuracy and weighted averaging.

Table 2. Hyperparameter Specifications

Hyperparameter	Value
Batch size	32
Early stopping	5
Epochs	100
Optimizer	adam
Learning rate	0.00001
Dropout	0.5

Thus, we delve into the consideration of other metrics, including precision, recall, and F1-Score, as well as overall accuracy. The macro-averaged recall takes the true positive rate into account across different classes, providing a more equitable inspection.

Suppose the quantities a , b , c , and d specify the true positive, true negative, false positive, and false negative, for our multi-class classification problem, respectively. In that case, the different metrics can be represented as follows:

$$Accuracy = \frac{a + b}{a + b + c + d} \quad (4)$$

$$Precision(Prec) = \frac{a}{a + c} \quad (5)$$

$$Recall(Rec) = \frac{a}{a + d} \quad (6)$$

$$F1 - Score(F1) = 2 \times \frac{Prec \times Rec}{Prec + Rec} \quad (7)$$

4.4 Simulations

In this work, for the assessment of our proposed network’s effectiveness, we conducted multiple simulations and discussed the results and their associated findings in this section.

Comparisons with Other Pre-trained CNN Models For our first set of simulations, we test with multiple pre-trained CNN models and report their accuracy, weighted & macro metrics for precision (prec), recall (rec), F1-Score (F1), respectively, in Table 3. Accordingly, we have selected CNN families that are diverse in terms of parameter count, for example, the parameter-heavy models, such as variations of ResNet, DenseNet, VGG, and NASNet, as well as lightweight ones like MobileNet, among others. From the table, we observe that the parameter-heavy models inadvertently achieved the best metric values, in the order of ResNets, followed by DenseNet, VGG, and others. However, the lightweight models maintain reasonable values, with the inception

Table 3. Comparisons among multiple Pre-trained Models

Model	Accuracy	Macro			Weighted		
		Prec	Rec	F1	Prec	Rec	F1
DenseNet169	78%	81%	80%	79%	81%	78%	78%
DenseNet201	79%	79%	81%	79%	79%	79%	78%
Xception	67%	68%	70%	67%	68%	67%	66%
VGG16	76%	76%	78%	77%	76%	76%	76%
VGG19	75%	75%	76%	75%	74%	75%	74%
ResNet50	79%	79%	80%	79%	79%	79%	78%
ResNet101	83%	83%	85%	83%	83%	83%	82%
ResNet152	80%	81%	81%	81%	80%	80%	80%
InceptionV3	66%	67%	68%	66%	67%	66%	64%
InceptionResNetV2	37%	27%	34%	29%	28%	37%	31%
MobileNet	73%	76%	76%	73%	77%	73%	72%
MobileNetV2	70%	71%	72%	70%	71%	70%	69%
NASNetMobile	63%	63%	65%	63%	62%	63%	61%
NASNetLarge	66%	74%	67%	68%	73%	66%	67%

family incurring the worst values, leading to ignoring them. With the compared ResNet family-based models performing the most sensible metrics, and ResNet101 achieving the best values, we have selected it as the base model for our experiments, as it obtains improvements of at least 3% over the second-best values for accuracy metrics. A similar trend is obtained when taking precision, recall, and F1 metrics as well, making it the default choice at this stage.

Table 4. Lentil Disease Detection Classification Report

Model	Ascochyta Blight			Lentil Rust			Normal			Powdery Mildew		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
DenseNet169	60%	83%	69%	86%	99%	92%	86%	67%	75%	93%	70%	80%
DenseNet201	81%	77%	79%	79%	96%	87%	80%	57%	67%	76%	95%	85%
Xception	61%	65%	63%	62%	93%	74%	72%	41%	52%	76%	81%	78%
VGG16	78%	62%	69%	77%	93%	84%	72%	62%	67%	80%	95%	87%
VGG19	70%	57%	63%	84%	83%	84%	67%	69%	68%	81%	94%	87%
ResNet50	87%	73%	79%	76%	92%	83%	73%	66%	69%	81%	89%	85%
ResNet101	81%	82%	82%	88%	93%	90%	81%	64%	72%	82%	99%	90%
ResNet152	82%	78%	80%	89%	83%	86%	77%	66%	71%	76%	98%	85%
InceptionV3	53%	54%	54%	85%	75%	79%	68%	43%	52%	64%	100%	78%
InceptionResNetV2	0%	0%	0%	25%	39%	30%	45%	78%	57%	38%	21%	27%
MobileNet	61%	88%	72%	77%	91%	83%	88%	43%	58%	78%	83%	80%
MobileNetV2	63%	69%	66%	74%	95%	83%	81%	47%	59%	65%	79%	71%
NASNetMobile	53%	47%	50%	74%	73%	73%	62%	43%	51%	63%	96%	76%
NASNetLarge	47%	85%	60%	95%	78%	86%	74%	55%	63%	79%	53%	63%

Next, we present the per-class classification report for different lentil disease categories in Table 4. From the table, it is interesting to observe that much of the variation in metric values across the models is evident for the normal category. While the lightweight MobileNet family achieves the best precision values, it suffers from the worst F1 values. Interestingly, only three models, namely, DenseNet169, ResNet101, and ResNet152, manage to overcome the 70% par values. This trend clearly suggests that the majority of the compared models are biased towards detecting diseased classes, potentially leading to the misclassification of normal samples as diseased, and suggesting over-sensitivity to features from diseased classes. Thus, to mitigate the aforementioned effect, we focus on the three models mentioned above for the remainder of this article. Next among the three, ResNet101 overpowers ResNet152 significantly with better F1-score values for all classes. Finally, although the DenseNet169 model achieves good values for lentil rust and the normal class, ResNet101 achieves almost comparable values for these classes while overpowering the remaining two categories, managing a $\sim 21\%$ improvement in precision for the Ascochyta Blight class. Given the upgrades, ResNet101 maintains its lead both at the overall level and on the per-class report as well.

Table 5. Empirical Layer Training Vs. Overall Metrics of ResNet101

Layer	Accuracy	Trainable Params(in Mil)	Macro			Weighted		
			Prec	Rec	F1	Prec	Rec	F1
5	81%	2.11	81%	83%	81%	82%	81%	81%
10	83%	5.52	83%	85%	83%	83%	83%	82%
15	82%	6.57	83%	84%	83%	83%	82%	82%
20	79%	9.98	80%	82%	79%	81%	79%	78%
25	84%	11.04	84%	86%	85%	84%	84%	84%
30	81%	15.5	82%	83%	81%	82%	81%	80%

Empirical Layer Re-training This section addresses the evaluation of layer freezing in relation to various performance metrics. Accordingly, we alter the number of layers at the end that are subjected to retraining, while keeping the rest of the layers frozen, and report their effect on performance. In accordance, we present the accuracy and the macro & weighted metrics along with the number of trainable parameters (measured in millions) in Table 5. As per the table, apart from the 20th entry, which acts as an anomaly, we observe that the layer freezing count in multiples of five results in improved metric values up to 25, following which it drops significantly (by $\sim 3\%$ for accuracy). Meanwhile, the 25-layer retraining achieves the best values for all metrics, but incurs a typical higher computational cost. Thus, the need arises to select the layer count that balances the computational cost and related metric values. We observe that with 10 layers fixed for training, we manage to obtain the second-best values across all layer counts while requiring a 50% reduction in parameter count. Although

the five-layer-based retraining ensures the least computational cost, it results in a 2% drop in accuracy, making it a suboptimal choice. Thus, rather than unthinkingly selecting the best metric value, our work presents a trade-off and selects the 10 layers as our optimal training for this work.

Table 6. Lentil Disease Detection Classification Report Among State-of-the-art

Model	Ascochyta Blight			Lentil Rust			Normal			Powdery Mildew		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
DenseNet201 [11]	70%	84%	77%	86%	100%	93%	94%	60%	73%	88%	100%	94%
ResNet101	81%	82%	82%	88%	93%	90%	81%	64%	72%	82%	99%	90%

Comparisons with state-of-the-art works Table 6 presents the per-class classification report between different state-of-the-art works and our proposed method. Interestingly, our method maintains a neck-and-neck fight for the normal class, where most other pre-trained models failed to score. While DenseNet201 [11] surpasses for the Lentil rust and powdery mildew classes, the improvements over our model are not significant margins. In contrast, ResNet101 gains at least $\sim 5\%$ for the Ascochyta blight class, with the precision gaining over $\sim 11\%$.

5 Conclusion & Future Works

This work centered on predicting lentil crop disease categories from a field-based, real-world dataset. With image datasets, popular strategies have been explored, including transfer learning with fine-tuning. While many literary works maintained a hardcoded approach for layer retraining, our approach experimentally validated and documented the trainable parameters while balancing, a feat often ignored. Compared to multiple pre-trained models, our model obtains overall improvements of $\sim 4\%$ while grabbing a significant $\sim 17\%$ for precision over state-of-the-art work on Ascochyta Blight disease. Therefore, fitted with balanced trainable parameters and comparable performance metrics, our work provides an experimental tuning for practical training.

Real-world smart farming applications require deployment at edge devices that are inherently low on storage and computational power. Using the parameter-heavy pre-trained models to deploy at the edge in their current condition is highly impractical. Therefore, for our future work, we will also explore the impact of compression techniques on reducing the model size while achieving comparable metric values.

References

1. Aktar-Uz-Zaman, M., Haque, M.A., Sarker, A., Rohman, M.M., Sarkar, D., Muzahid-E-Rahman, M., Haque, R.B., Akhter, M.S., Agrawal, S.K.: Screening of

- lentil genotypes against stemphylium blight disease and molecular identification of causal organism. *Scientific Reports* **15**(1), 32318 (2025)
2. Aravind, K., Raja, P., Mukesh, K., Anirudh, R., Ashiwin, R., Szczepanski, C.: Disease classification in maize crop using bag of features and multiclass support vector machine. In: 2018 2nd international conference on inventive systems and control (ICISC). pp. 1191–1196. IEEE (2018)
 3. Ashurov, A.Y., Al-Gaashani, M.S., Samee, N.A., Alkanhel, R., Atteia, G., Abdallah, H.A., Saleh Ali Muthanna, M.: Enhancing plant disease detection through deep learning: a depthwise cnn with squeeze and excitation integration and residual skip connections. *Frontiers in Plant Science* **15**, 1505857 (2025)
 4. Bayaa, B., Erskine, W.: Lentil pathology. *The Pathology of Food and Pasture Legumes* (D. Allen, J. Lenné, eds.). Commonwealth Agricultural Bureaux International, Slough, Berkshire, UK pp. 422–471 (1998)
 5. Behera, S., Misra, R.: A multi-model data-fusion based deep transfer learning for improved remaining useful life estimation for iiot based systems. *Engineering Applications of Artificial Intelligence* **119**, 105712 (2023)
 6. Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L.: Imagenet: A large-scale hierarchical image database. In: 2009 IEEE conference on computer vision and pattern recognition. pp. 248–255. Ieee (2009)
 7. Food, of the United Nations, A.O.: World food and agriculture-statistical yearbook 2020. Food and Agriculture Organization of the United Nations (2020)
 8. Food and Agriculture Organization of the United Nations (FAO): Faostat database. <http://faostat.fao.org> (2018), accessed: 2025-10-30
 9. Islam, M.A., Islam, S.M.A., Sathi, M.A., et al.: Identification of lentil varieties/lines resistant to stemphylium blight considering disease reaction and yield. *Malays J Sustain Agric* **4**, 22–25 (2020)
 10. Kumar, S., Barpete, S., Kumar, J., Gupta, P., Sarker, A.: Global lentil production: constraints and strategies. *SATSA Mukhapatra-Annual Technical* **17**, 1–13 (2013)
 11. Mahamud, E., Assaduzzaman, M., Sharmin, S.: Lentil plant disease and quality assessment: A detailed dataset of high-resolution images for deep learning research. *Data in Brief* **58**, 111224 (2025)
 12. Selvaraj, K.M., Gnanagurusubbiah, S., Roy, R.R.R., Balu, S., et al.: Enhancing skin lesion classification with advanced deep learning ensemble models: a path towards accurate medical diagnostics. *Current Problems in Cancer* **49**, 101077 (2024)
 13. Sen Gupta, D., Thavarajah, D., Knutson, P., Thavarajah, P., McGee, R.J., Coyne, C.J., Kumar, S.: Lentils (*lens culinaris* l.), a rich source of folates. *Journal of agricultural and food chemistry* **61**(32), 7794–7799 (2013)
 14. Shah, N., Jain, S.: Detection of disease in cotton leaf using artificial neural network. In: 2019 amity international conference on artificial intelligence (AICAI). pp. 473–476. IEEE (2019)
 15. Sinha, J., Singh, A.: Effect of environment on the development and spread of stemphylium blight of lentil. (1993)
 16. Thai, H.T., Le, K.H.: Mobileh-transformer: Enabling real-time leaf disease detection using hybrid deep learning approach for smart agriculture. *Crop Protection* **189**, 107002 (2025)
 17. Varma, T., Mate, P., Azeem, N.A., Sharma, S., Singh, B.: Automatic mango leaf disease detection using different transfer learning models. *Multimedia Tools and Applications* **84**(11), 9185–9218 (2025)
 18. Weiss, K., Khoshgoftaar, T.M., Wang, D.: A survey of transfer learning. *Journal of Big data* **3**(1), 9 (2016)